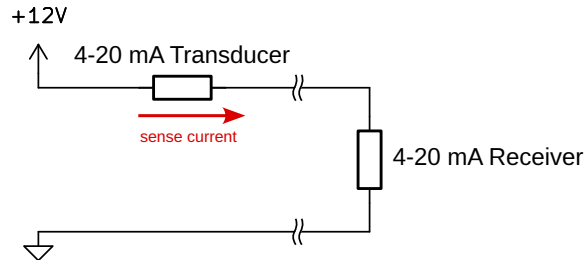


Homework 3

Due February 11th

Problems:

A common method for transmitting sensor information in an industrial or laboratory setting is to use what is known as a 4-20 mA current loop, which is shown in a simple fashion below.



In this process, a sensor and current transmitter (the “transducer”) are powered by a voltage source, e.g., 12 V. The transducer converts the sensed process (e.g., [temperature](#) or [pressure](#)) to an output current which is fed to a receiver in the same loop. The receiver converts the current back into a useable form (e.g., voltage, digital value). The use of current to send information lends itself well to electromagnetically noisy environments as a current receiver is inherently a low-impedance input, which tends to remove low-energy noise sources. Further, the transducer and receiver may be separated by a large distance: the current must be consistent around the loop regardless of any voltage loss due to resistance in cabling. Powering of the transducer may be completely done via the current loop itself, allowed via the 4 mA output offset. The receiver is typically assumed to be powered locally.

1. Considering the 4-20 mA loop as described, do the following:
 - (i) Design the transmitter portion of the transducer using one of the topologies discussed in Chapter 2 or the lectures*. The transmitter must convert a sensor input voltage of -2 V to 2 V such that 4 mA corresponds to -2 V and 20 mA corresponds to 2 V. Assume ideal amps without saturation and that you may use as many amps as necessary. Any resistors selected should be reasonably valued. As opposed to the diagram as shown, you may assume both ground and 12 V, and any other voltage source desired is available in the transmitter; however, *floating DC supplies are not allowed*. Calculate the voltage compliance of the designed circuit. Hint: Think about how the remainder of the loop is connected to the transmitter in determining its topology.
 - (ii) Design the 4-20 mA receiver circuit such that it converts the transmitted 4-20 mA current to a 0-5 V range for connection to an ADC. Again use ideal assumptions and that you may use as many amps as necessary. The receiver *must be compatible* with the transmitter as designed in the previous step.
 - (iii) Verify your design within LTSpice (preferred) or another simulation package. An ideal op amp is available within LTSpice as component UniversalOpAmp within the OpAmps component category. **Provide image captures of both the schematic and simulation results, showing both the input and output voltages and current transmitted.** The simulation should show the circuits work over the whole input voltage range. More information about LTSpice is at the end of this document. See me or TA in office hours if you need help using LTSpice.
2. Book Problem 2.21, (-4050: part a only, -6050: both a and b). The target solution for i_o in the book is wrong, it should be: $i_o = A_2 v_2 - A_1 v_1 - v_L / R_o$ instead. Find the condition where $R_o \rightarrow \infty$ which also causes $A_2 = A_1 = A$. Assume ideal op amps.

*This is usually not how you would construct the transmitter for the actual product, but good enough for this homework!

3. Book Problem 2.25. Assume ideal op amps.
4. Book Problem 2.34. Assume ideal op amps.
5. -6050 Only: Book Problem 2.39. Note that the problem defines A_{CM} and A_{DM} in terms of short-circuit load current, implying that your analysis can be done assuming $R_L = 0 \Omega$.

LTSpice Links/Info:

- LTSpice homepage (with download links): <https://www.analog.com/en/design-center/design-tools-and-calculators/simulator.html>
 - LTSpice has installers for both Windows and MacOS. LTSpice runs great in Linux under [Wine](#) using the Windows installer.
- Getting Started Videos: <https://www.analog.com/en/education/education-library/videos/video-series/ltspice-getting-started-tutorial.html>
- An older tutorial PDF (slides 14-38 applicable): <https://www.analog.com/media/en/simulation-models/spice-models/ltspicegettingstartedguide.pdf?modelType=spice-models>
- Keyboard shortcuts: <https://www.analog.com/media/en/news-marketing-collateral/solutions-bulletins-brochures>